Strategic Analysis of Computer Vision Data Ecosystems for the 'Fessi' Waste Management Knowledge Graph

1. Introduction: The Convergence of Semantic Knowledge and Visual Perception in Waste Management

The deployment of 'Fessi'—a knowledge graph-driven chatbot designed for a German university waste management context—represents a significant evolution in automated environmental compliance systems. While traditional waste management relies on static signage and user volition, Fessi aims to bridge the cognitive gap between complex regulatory frameworks (*Kreislaufwirtschaftsgesetz*) and the instantaneous, chaotic reality of waste disposal on a busy campus. The success of this system hinges not merely on the semantic richness of its Knowledge Graph (KG) but, crucially, on its visual perception capabilities. The user query necessitates a rigorous discovery and evaluation of Computer Vision (CV) datasets that can empower Fessi to interpret the visual world—classifying waste items, estimating volumes, detecting contamination, and ensuring laboratory safety through hazardous material recognition.

This report provides an exhaustive analysis of the data landscape required to train such a vision system. The challenge is multifaceted: the system must transition from identifying idealized studio-quality objects to recognizing deformed, occluded, and contaminated waste in "wild" environments ranging from lecture halls to chemical laboratories. Furthermore, the visual domain must be strictly aligned with the specificities of the German recycling ecosystem, encompassing the *Pfand* (deposit) system, the distinct separation of lightweight packaging (*Gelber Sack*), paper (*Blaue Tonne*), and hazardous laboratory waste.

Current literature and available repositories reveal a dichotomy in waste datasets: a prevalence of small, clean, academic datasets suitable for benchmarking, contrasted with a growing but fragmented collection of industrial-grade and "in-the-wild" datasets that reflect true operational complexity. The integration of these disparate sources into a cohesive training pipeline for Fessi demands a strategic evaluation of their size, annotation granularity, environmental context, and licensing compatibility. This report dissects over thirty distinct data sources, categorizing them into functional domains—Domestic, Industrial, Construction, Hazardous, and German-specific—to provide a roadmap for constructing the visual cortex of the Fessi ecosystem.

2. General Waste and Domestic Trash Classification

The foundational capability of Fessi lies in its ability to classify common domestic waste items—packaging, food scraps, paper, and general refuse—generated by the student and faculty population. This domain serves as the primary interface for user interaction, where photo-based queries regarding "which bin does this go in?" are most frequent. The landscape of available data for this task is characterized by a spectrum ranging from highly controlled "studio" datasets to complex "in-the-wild" collections.

2.1 Studio and Controlled Environment Datasets

Controlled environment datasets provide the baseline feature representations necessary for any waste classification model. By isolating objects against neutral backgrounds, these datasets allow Convolutional Neural Networks (CNNs) and Transformers to learn the intrinsic visual properties of materials—the specular highlights of PET plastic, the fibrous texture of cardboard, or the metallic sheen of aluminum—without the confounding variables of background clutter or lighting variations.

2.1.1 TrashNet: The Academic Benchmark

TrashNet [1] remains the ubiquitous starting point for waste classification research. Developed by researchers at Stanford University, the dataset consists of 2,527 images categorized into six classes: glass, paper, cardboard, plastic, metal, and trash. The images are captured against a white board, ensuring high object salience and minimal background noise.

While TrashNet is instrumental for initial model benchmarking and validating transfer learning architectures (such as ResNet or DenseNet), its utility for a deployed system like Fessi is limited by its lack of environmental diversity. The "studio" nature of the data means that models trained solely on TrashNet often suffer from significant performance degradation when presented with real-world images where lighting is uncontrolled and backgrounds are complex. Furthermore, the limited taxonomy of six classes fails to capture the granularity required for German waste separation rules. For instance, the class "plastic" in TrashNet does not distinguish between a PET bottle (which may carry a *Pfand* deposit) and a polystyrene yogurt cup (which belongs in the *Gelber Sack*). Despite these limitations, TrashNet's MIT license [2] and standardized format make it a valuable resource for pre-training early layers of the vision model before fine-tuning on more complex datasets.

2.1.2 Garbage Classification V2: Expanding the Taxonomy

Addressing the class limitations of TrashNet, the Garbage Classification V2 dataset [3] represents a significant expansion in both volume and taxonomy. With approximately 15,150 images distributed across 12 classes, this source provides the breadth necessary to handle a wider array of campus waste. The inclusion of a "Biological" class [3] is particularly relevant for the university context, enabling Fessi to advise on the disposal of food scraps into the

Biotonne (organic waste bin), a critical component of German waste diversion strategies.

Crucially, this dataset introduces a "Battery" class. In the context of facility management, batteries represent a severe fire hazard if disposed of in general waste or paper bins. A vision model capable of explicitly detecting batteries allows Fessi to intervene with high-priority safety warnings, directing users to specific collection points rather than standard bins. The dataset also separates "Paper" and "Cardboard," a distinction that aligns with volume reduction strategies; Fessi can advise users to flatten cardboard boxes to maximize bin capacity, a nuance lost if the classes are merged.

While the dataset is sourced from Kaggle and presents a mix of image qualities, its size allows for the training of deeper architectures like EfficientNet-B0 or ResNeXt-50 [5] without immediate overfitting. The licensing is generally open (often CC0 or similar on Kaggle), though verification of individual image provenance is recommended for strict academic compliance.

2.2 In-the-Wild and Litter Datasets

To bridge the "domain gap" between training data and the chaotic reality of user-submitted photos, Fessi must be trained on images captured in the wild. Users will likely photograph waste held in their hands, lying on a cafeteria table, or scattered on the ground. "In-the-wild" datasets capture these unconstrained scenarios, including varying lighting conditions, occlusions, and diverse backgrounds.

2.2.1 TACO: Trash Annotations in Context

The Trash Annotations in Context (TACO) dataset [2] stands out as the premier resource for litter detection and segmentation in the wild. Unlike the bounding box annotations found in many datasets, TACO provides high-quality segmentation masks (polygons) for its objects. This granular annotation style is critical for the "contamination detection" requirement of the user query. By understanding the precise shape and boundaries of an object, the vision system can better identify if a recyclable item is soiled, crumpled, or partially obscured by non-recyclable contaminants.

TACO features 60 categories mapped to 28 super-categories, offering a hierarchical ontology that integrates seamlessly with a Knowledge Graph structure. For example, specific items like "Drink carton" or "Crisp packet" can be mapped to parent nodes like Composite_Packaging or Multi-layer_Plastic within Fessi's graph, enabling precise disposal advice for the *Gelber Sack*. The dataset covers diverse environments including forests, roads, and beaches, providing the visual robustness needed for a campus-wide system that includes outdoor areas. Released under an MIT license [2], TACO allows for unrestricted academic and commercial use, making it a cornerstone dataset for Fessi.

2.2.2 MJU-Waste: The Depth Dimension

A unique requirement of the user query is "volume estimation." Standard RGB images suffer

from scale ambiguity—a small object close to the camera can appear identical in size to a large object further away. The MJU-Waste dataset [7] addresses this by providing co-registered RGB and Depth (RGB-D) data.

Comprising 2,475 image pairs, MJU-Waste allows Fessi to be trained not just to recognize *what* an object is, but *how big* it is. By leveraging the depth channel during training, the model can learn to infer volume from monocular images (or utilize depth data if the user's device supports it, such as modern iPhones with LiDAR). This capability is vital for facility management: Fessi can analyze a photo of a bin and estimate its fill level, or assess a pile of bulky waste to determine if it requires a special pickup service (*Sperrmüll*). The dataset includes instance segmentation masks [9], enabling the separation of overlapping objects in a pile, further enhancing volume estimation accuracy.

2.3 Integration and Strategic Synthesis of Domestic Waste Data

The integration of these domestic waste datasets requires a layered approach within Fessi's architecture. The "Studio" datasets (TrashNet, Garbage Classification V2) should be employed to train the feature extractors of the model, ensuring it has a robust understanding of material textures and basic shapes. The "Wild" datasets (TACO) should then be used for fine-tuning the model's localization and detection heads, teaching it to find these objects amidst the clutter of a university campus.

The segmentation masks provided by TACO and MJU-Waste offer a pathway to advanced "Contamination Detection." By analyzing the pixel-level coherence of an object, Fessi can detect anomalies—such as food residue on a pizza box or liquid remaining in a bottle. This visual analysis triggers specific logic in the Knowledge Graph: if "contamination" is detected, the disposal advice shifts from "Blue Bin" (Paper) to "Black Bin" (Residual Waste), thereby preserving the purity of the recycling stream.

Dataset Name	Primary Domain	Size (Images)	Annotation Type	Key Strengths for Fessi	License
TrashNet	Studio / Benchmark	2,527	Classification	Baseline texture learning; Standard benchmark	MIT [2]
Garbage Class. V2	General / Domestic	15,150	Classification	12 Classes including	Open [3]

				Bio & Battery; Large scale	
TACO	Litter / Wild	1,500+ (High object count)	Segmentation (Polygons)	Hierarchical classes; Contamination detection; Wild context	MIT [2]
MJU-Waste	Volume / Depth	2,475 (RGB-D pairs)	Segmentation + Depth	Volume estimation; Shape analysis; RGB-D training	MIT [7]

Table 1: *Strategic comparison of Domestic Waste Datasets, highlighting their specific utility for the Fessi Knowledge Graph project.*

3. Industrial and Conveyor Belt Datasets for Contamination and Sorting

While domestic datasets address the "user query" aspect of Fessi, the requirement for "recycling bin sorting" and "contamination detection" necessitates data that mimics the conditions of waste aggregation. Industrial datasets, typically captured on moving conveyor belts in Material Recovery Facilities (MRFs), provide a high-density, high-occlusion visual environment that is invaluable for training robust detection models.

3.1 ZeroWaste: Handling Extreme Clutter and Deformation

The ZeroWaste dataset [10] represents a quantum leap in the complexity of waste detection data. Collected from a real operational MRF in the United States, specifically from a high-quality paper sorting line, it captures waste in its most chaotic state. The dataset focuses on "extreme clutter" and highly deformable objects, addressing one of the most persistent challenges in computer vision: recognizing objects that have lost their canonical shape.

For Fessi, the "Soft Plastic" class within ZeroWaste is of particular strategic importance. In the German recycling system, distinguishing between rigid plastics (like bottles and trays) and

soft plastics (films, foils, wrappers) is often a source of confusion for users and a technical challenge for automated systems. Soft plastics are highly deformable, often translucent, and prone to crumpling, making them difficult to detect with models trained on rigid objects. ZeroWaste provides robust training examples for these materials [12], enabling Fessi to correctly identify a crumpled plastic wrapper as "Soft Plastic" and direct it to the *Gelber Sack*.

Furthermore, the dataset's focus on occlusion—where objects are partially hidden by others—directly supports the "contamination detection" requirement. A model trained on ZeroWaste learns to spot a small piece of metal or plastic peeking out from under a pile of paper. This capability allows Fessi to analyze photos of a student's recycling bin and flag "contaminants" that might ruin the batch, advising the user to remove them before disposal. The dataset is provided in the standard MS COCO format [13], facilitating easy integration into modern object detection pipelines like YOLO or Detectron2.

3.2 RealWaste: Landfill and Reception Analysis

Complementing the sorting-line perspective of ZeroWaste, the RealWaste dataset [14] captures images at the "point of reception" in a landfill. This dataset is characterized by high-resolution imagery and a taxonomy that includes "Textile Trash" and "Vegetation."

The "Textile" class is a valuable addition for a university campus context. Old clothing and fabrics are frequently mishandled, often ending up in general waste or contaminating recycling bins. By explicitly recognizing textiles, Fessi can guide users to the *Altkleidercontainer* (clothing donation bins) often located near campus residences, promoting circularity. The dataset's "Vegetation" class also reinforces the separation of green waste, supporting campus groundskeeping efforts.

3.3 SpectralWaste: The Future of Material Recognition

Although the primary focus of the user query is on standard computer vision, the SpectralWaste dataset [15] introduces the dimension of Hyperspectral Imaging (HSI) alongside RGB. While Fessi's end-users will likely rely on smartphone cameras (RGB), the insights from SpectralWaste are valuable for feature engineering. HSI data reveals that certain materials, like Polypropylene (PP) and Polyethylene Terephthalate (PET), have distinct spectral signatures despite looking identical in RGB. By studying the correlations between HSI and RGB features in this dataset, researchers can identify subtle visual cues—such as specific texture-color combinations—that serve as proxies for material composition in standard images. This "third-order" insight can enhance the accuracy of Fessi's material recognition logic even without hyperspectral sensors.

4. Construction and Demolition (C&D) Waste Identification

University campuses are dynamic environments frequently undergoing renovation, repair, and construction. The waste generated from these activities—rubble, wood, insulation, and piping—is subject to strict disposal regulations and high disposal costs. Misclassifying C&D waste as municipal waste can lead to heavy fines and operational disruptions. Therefore, Fessi requires a dedicated module for C&D identification.

4.1 CODD: The Construction and Demolition Waste Object Detection Dataset

The CODD dataset [16] is the primary resource identified for this domain. Captured at a recycling facility in Cyprus, it offers a rigorous collection of 3,129 high-resolution images with over 16,000 annotated samples. The dataset's taxonomy of ten distinct classes—Bricks, Concrete, Tiles, Wood, Pipes, Plastics, General Waste, Foaming Insulation, Stones, and Plaster Boards—provides the granularity necessary for regulatory compliance.

In the German context, the distinction between "Bricks" (*Ziegel*) and "Plaster Boards" (*Gips*) is critical. Gypsum-based materials must often be separated from mineral rubble (*Bauschutt*) due to sulfate contamination risks which can compromise the quality of recycled concrete. A model trained on CODD can enforce this separation logic within Fessi, advising users to segregate plasterboard into dedicated containers. Additionally, the detection of "Foaming Insulation" is vital, as older insulation materials can be hazardous (containing HBCD or asbestos), triggering a safety protocol in the chatbot.

The annotations in CODD are provided in PASCAL VOC XML format, including both bounding boxes for detection and polygons for segmentation. The polygon annotations allow for precise area calculation of waste piles. When combined with reference objects or depth data (from MJU-Waste training), this enables Fessi to estimate the volume of construction debris—a key requirement for ordering the correct size of skip (*Container*) for a renovation project.

4.2 RWTH Aachen RCM: Material Science in Waste

Originating from RWTH Aachen University, the RCM (Recycled Concrete) dataset [18] offers a highly specialized, academic perspective on C&D waste. Unlike general detection datasets, RCM focuses on the particle size distribution (PSD) and shape analysis of recycled aggregates.

This dataset enables "Deep Material Analysis." It allows Fessi to move beyond simple classification (e.g., "this is rubble") to qualitative assessment (e.g., "this rubble consists of 70% concrete and 30% brick with an average grain size of 50mm"). While this level of detail might exceed the needs of a typical student, it is invaluable for facility managers and sustainability officers auditing the campus's construction recycling rates. It supports the circular economy goal of verifying the quality of secondary raw materials directly on site. The dataset's provenance from a German technical university ensures its alignment with local

material standards and research rigor.

5. Hazardous, Laboratory, and Medical Waste: A Critical Safety Layer

For a university with active research faculties in chemistry, biology, and medicine, the correct identification of hazardous waste is a non-negotiable safety requirement. Standard waste datasets completely lack these categories, making the integration of specialized hazardous waste data a priority for Fessi.

5.1 LabPics: Chemistry and Medical Contexts

The LabPics dataset [19] is the definitive resource for the laboratory context. It contains images of materials inside glassware (beakers, flasks, funnels) and medical settings. The dataset goes beyond object detection to segment the materials *inside* the vessels, identifying phases such as liquids, solids, and foams.

This granularity enables sophisticated safety logic in Fessi. If a user photographs a beaker containing a clear liquid, Fessi can identify it as "Laboratory Glassware with Liquid Phase." The Knowledge Graph then triggers a query regarding the nature of the liquid. Instead of suggesting a glass recycling bin (which would be dangerous if the liquid were a solvent), Fessi can advise the user to empty the vessel into the appropriate chemical waste canister before cleaning and disposal. The dataset also helps distinguish between "lab glass" (borosilicate, which melts at different temperatures and contaminates recycling) and "beverage glass" (soda-lime), ensuring that lab equipment never enters the municipal glass stream (*Altglascontainer*).

5.2 Hazmat Placards and GHS Symbols

The recognition of warning signage is a fundamental safety layer. The Hazmat Placard Image Dataset [22] and collections of GHS (Globally Harmonized System) hazard symbols [23] provide the data necessary to train a "Red Flag" detector.

Fessi's vision pipeline should be architected to scan every uploaded image for these pictograms—Explosive, Flammable, Corrosive, Biohazard, Radioactive—*before* attempting any general waste classification. The detection of a GHS symbol acts as a hard override, aborting standard disposal advice and triggering immediate safety protocols (e.g., "Biohazard detected: Do not touch. Contact the Safety Officer immediately."). This preemptive safety layer protects both the user and the waste management staff from exposure to dangerous materials.

5.3 Biomedical Waste: Sharps and Clinical Safety

In university hospitals and biology departments, biomedical waste such as syringes, scalpels,

and soiled dressings is common. Datasets like "Biomedical Waste Classification" [25] and the medical subset of "TrashBox" [26] provide images of these items in clinical contexts.

The distinction of "Sharps" (syringes, needles) is particularly critical. These items require disposal in puncture-proof containers (*Kanülenabwurfbehälter*). A generic waste classifier might misidentify a syringe as a "plastic pen" or "tube," leading to a needlestick injury risk for cleaning staff. By training on specialized biomedical datasets, Fessi can achieve high recall for sharps, ensuring they are strictly segregated from general trash streams.

6. The German Context: Localization Strategies for Pfand and Packaging

A generic waste classifier, no matter how accurate, will fail in Germany if it does not understand the local "rules of the game"—specifically the *Pfand* (deposit) system and the dual system for packaging waste. Fessi must be localized to navigate these uniquely German frameworks.

6.1 The Pfand System: Visual Turing Test for Recycling

Germany's deposit system differentiates between single-use (*Einweg*) and multi-use (*Mehrweg*) beverage containers, both of which carry a monetary deposit (0.08€ to 0.25€) and must be returned to retailers, not thrown in bins. The visual marker for this system is the DPG logo (Deutsche Pfandsystem GmbH) or specific bottle shapes and wear patterns associated with multi-use glass.

To automate this, Fessi requires a specialized "Logo Detector." Datasets such as "thats pfand again" [27] and "bank bottle" [28] on platforms like Roboflow are specifically annotated for bottle vending machines. Furthermore, the **Reverse Vending Machine (RVM) Dataset** [29] provides high-quality images of bottles and cans from top-down and front-facing perspectives, mimicking the view of a machine verifying an item.

By training on this data, Fessi can implement a "Pfand Check" logic:

1. **Detection:** Is the object a bottle or can?
2. **Verification:** Does it bear the DPG logo or resemble a standard *Mehrweg* bottle?
3. **Advice:** If yes, "Return to supermarket for deposit." If no, "Dispose in Yellow Bin (Plastic/Metal) or Glass Bin."

6.2 The "Gelber Sack" and Packaging Identification

The *Gelber Sack* (Yellow Bag) is destined exclusively for lightweight packaging (LWP)—yogurt cups, composite cartons (TetraPak), plastic films, and cans. It explicitly excludes non-packaging plastics (like a plastic toy or toothbrush), creating a complex decision

boundary for users.

While no massive public dataset exists solely for "German Yellow Bag contents," proxies like the **WaDaBa** dataset [31] (from Poland, which shares similar plastic streams) and **RWTH Aachen's NIR-MFCO** [32] provide relevant data. The RWTH dataset is particularly valuable as it analyzes post-consumer plastics at different material flow compositions, effectively modeling the contents of a Yellow Bag.

Fessi must be trained to recognize the visual cues of packaging: logos, barcodes, and specific material textures (foils, laminates). The "Soft Plastic" class from the ZeroWaste dataset is crucial here, as it represents the films and wrappers that make up a significant volume of the *Gelber Sack*. Additionally, detecting the "Grüner Punkt" (Green Dot) logo [33], while no longer legally mandatory for disposal, remains a strong heuristic for packaging waste that can be leveraged by the vision system.

6.3 Bin Classification and Context

Finally, Fessi needs to recognize the bins themselves to verify user actions. German recycling bins follow a color code: Blue (Paper), Yellow (Packaging), Brown/Green (Bio), and Black/Grey (Residual). Datasets identifying these bin types [34] allow Fessi to perform "Context Checks." If a user uploads a photo of a pizza box hovering over a Blue Bin, Fessi can detect the context and intervene: "Stop! Greasy pizza boxes belong in the Black Bin, not the Blue Bin."

7. Advanced Capabilities: Volume Estimation and Contamination Detection

The user query specifies a need for "volume estimation" and "contamination detection," capabilities that transform Fessi from a passive advisor into an active facility management tool.

7.1 Volume Estimation via RGB-D and Segmentation

To estimate volume, Fessi must move from 2D pixel area to 3D spatial understanding. The **MJU-Waste** dataset [7] is the key enabler here. By training on its paired RGB and Depth images, the model learns to correlate visual features (shading, perspective) with depth.

- **Mechanism:** Fessi segments the waste pile using a model trained on TACO or CODD (providing pixel area). It then infers the average depth of the pile using the depth features learned from MJU-Waste.
- **Calculation:** $Volume \approx Area \times AverageDepth$.
- **Application:** This allows the university to monitor bin fill rates autonomously. A photo of a bin can yield a "70% Full" metric, triggering a cleaning request only when necessary.

7.2 Contamination Detection via Segmentation and Clutter Analysis

Contamination—the presence of non-recyclables in a recycling stream—is a major cost driver. The **ZeroWaste** dataset [10] provides the training ground for this. Its images of extreme clutter allow the model to learn to detect "foreign objects" in a homogeneous stream.

- **Mechanism:** If the user claims to be disposing of "Paper," Fessi analyzes the image for anomalies. Using segmentation masks from TACO, it checks for objects with shapes or textures inconsistent with paper (e.g., the round rim of a coffee cup or the glint of a metal can).
- **Logic:** The detection of such an anomaly triggers a "Contamination Alert," prompting the user to remove the offending item before disposal.

8. Model Architectures and Integration Strategy

The selection of model architecture determines the speed, accuracy, and deployment feasibility of the system.

8.1 YOLOv8 / YOLOv11: The Real-Time Workhorse

The YOLO (You Only Look Once) family of models [5] is the industry standard for real-time object detection and is heavily represented in the research literature.

- **Advantage:** YOLO offers an optimal trade-off between inference speed and accuracy, crucial for a chatbot where latency impacts user experience.
- **Implementation:** Fessi should utilize **YOLOv8-seg** (Instance Segmentation). This variant provides both bounding boxes (for counting) and polygon masks (for shape/contamination analysis).
- **Pre-training:** Hugging Face repositories [38] offer YOLO models pre-trained on waste datasets, providing a warm-start for training.

8.2 Hierarchical Ensemble Approach

Given the complexity of the domain, a single model is insufficient. Fessi should employ a hierarchical ensemble:

1. **The Gatekeeper (YOLOv8):** A fast, generalist model scans the image for Hazmat symbols (GHS) and broad categories (Bottle, Box, Bag).
2. **The Specialist (Fine-tuned Heads):**
    - If "Bottle" is detected $\rightarrow$ Activate **Pfand-Net** (trained on RVM data) to check for logos.
    - If "Rubble" is detected $\rightarrow$ Activate **CODD-Net** (trained on C&D data) to distinguish brick from concrete.
    - If "Glassware" is detected $\rightarrow$ Activate **Lab-Net** (trained on LabPics) to check for chemical residues.

3. **The Semantic Mapper:** The outputs of these models (e.g., Class: "PET_Bottle", Condition: "Crushed", Logo: "None") are passed to the Knowledge Graph, which queries the regulations to generate the final advice: "Dispose in Yellow Bin."

9. Comprehensive Source Documentation

The following table summarizes the key datasets identified for the Fessi project, categorizing them by their primary utility and licensing status.

ID	Dataset Name	Primary Domain	Source Ref	Coverage & Quality	License	Integration Note
D1	**TACO**	Litter / General	[2]	High-quality segmentation; Wild context; 60 classes.	MIT	Critical for litter & contamination. Use masks for shape analysis.
D2	**ZeroWaste-f**	Industrial / Sorting	[10]	Extreme clutter; Soft plastic class; Conveyor view.	CC BY 4.0	Essential for "Soft Plastic" (Gelber Sack) & clutter handling.
D3	**MJU-Waste**	Volume / Depth	[7]	RGB-D pairs; Indoor/Outdoor.	MIT	Key for "Volume Estimation" feature. Train depth estimator here.

D4	**CODD**	Construction (C&D)	[17]	10 classes (Brick, Concrete, Pipes); High Res.	CC BY 4.0	Activate for renovation queries. Maps to "Bauschutt" regulations.
D5	**LabPics**	Laboratory	[19]	Glassware, Liquid/Solid phases; Medical.	CC BY 4.0	Vital for lab safety. Detects "full" vessels.
D6	**Garbage Class V2**	Domestic	[3]	15k images; 12 classes (Bio, Battery).	Open	Source for "Battery" & "Bio" training. Good baseline size.
D7	**RVM Dataset**	Pfand / Bottles	[29]	Top/Front views; Fraud detection context.	MIT	Foundation for "Pfand Check" logic.
D8	**Hazmat Placards**	Safety	[22]	GHS Symbols; Transport signs.	Open	The "Red Flag" detector. Must run on every image.

D9	**RWTH RCM**	Material Science	[18]	Particle size; Concrete vs Brick analysis.	Academic	Advanced material analysis for circular economy reporting.

10. Conclusion and Strategic Recommendations

The visual intelligence of the 'Fessi' chatbot cannot be built upon a single, generic dataset. The requirement to operate within the regulatory framework of a German university—navigating the *Pfand* system, hazardous laboratory protocols, and strict separation of C&D waste—demands a composite data strategy.

It is recommended that the development team adopts a **multi-model architecture**. A core **YOLOv8** model, pre-trained on **Garbage Classification V2** and fine-tuned on **TACO**, should serve as the general interaction layer. This must be augmented by specialized modules: a **Safety Layer** trained on **Hazmat** and **LabPics** data that preemptively scans for danger; a **Localization Layer** trained on **RVM** and **ZeroWaste** data to handle German packaging nuances; and a **Facility Management Layer** leveraging **MJU-Waste** and **CODD** for volume and debris analysis.

By synthesizing these diverse data sources into a coherent visual pipeline, Fessi can evolve from a simple query tool into a robust, visually aware agent that actively supports the university's sustainability and safety goals. The mapping of these visual recognitions to the semantic depth of the Knowledge Graph will ensure that the advice given is not just visually accurate, but legally and contextually correct.

Works cited

1. Towards sustainable solutions: Effective waste classification framework via enhanced deep convolutional neural networks | PLOS One, accessed January 11, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0324294
2. AgaMiko/waste-datasets-review: List of image datasets with any kind of litter, garbage, waste and trash - GitHub, accessed January 11, 2026, https://github.com/AgaMiko/waste-datasets-review
3. Garbage Classification (12 classes) - Kaggle, accessed January 11, 2026, https://www.kaggle.com/datasets/mostafaabla/garbage-classification
4. Garbage Dataset - Kaggle, accessed January 11, 2026, https://www.kaggle.com/datasets/sumn2u/garbage-classification-v2
5. Health Care Waste Classification Using Deep Learning Aligned with Nepal's Bin

Color Guidelines - arXiv, accessed January 11, 2026, https://arxiv.org/html/2508.07450v1
6. jeremy-rico/litter-detection: Litter detection with YOLOv8 and TACO - GitHub, accessed January 11, 2026, https://github.com/jeremy-rico/litter-detection
7. realwecan/mju-waste: The MJU-Waste dataset from our ... - GitHub, accessed January 11, 2026, https://github.com/realwecan/mju-waste
8. MJU-Waste - OpenDataLab, accessed January 11, 2026, https://opendatalab.com/OpenDataLab/MJU-Waste/download
9. MJU-Waste - Dataset Ninja, accessed January 11, 2026, https://datasetninja.com/mju-waste
10. ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes, accessed January 11, 2026, http://ai.bu.edu/zerowaste/
11. ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes - CVF Open Access, accessed January 11, 2026, https://openaccess.thecvf.com/content/CVPR2022/papers/Bashkirova_ZeroWaste_Dataset_Towards_Deformable_Object_Segmentation_in_Cluttered_Scenes_CVPR_2022_paper.pdf
12. zerowaste Object Detection Dataset by Boston University, accessed January 11, 2026, https://universe.roboflow.com/boston-university/zerowaste-tundo
13. [CVPR 2022] ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes - GitHub, accessed January 11, 2026, https://github.com/dbash/zerowaste
14. RealWaste - UCI Machine Learning Repository, accessed January 11, 2026, https://archive.ics.uci.edu/dataset/908/realwaste
15. SpectralWaste Dataset: Multimodal Data for Waste Sorting Automation - arXiv, accessed January 11, 2026, https://arxiv.org/html/2403.18033v1
16. Construction and Demolition Waste Object Detection Dataset - Mendeley Data, accessed January 11, 2026, https://data.mendeley.com/datasets/24d45pf8wm/2
17. Construction and Demolition Waste Object Detection Dataset ..., accessed January 11, 2026, https://data.mendeley.com/datasets/wds85kt64j/3
18. Deep learning-based prediction of particle size distributions in construction and demolition waste recycling using convolutional - RWTH Publications, accessed January 11, 2026, https://publications.rwth-aachen.de/record/1005633/files/1005633.pdf
19. LabPics Chemistry / LabPics Medical - Kaggle, accessed January 11, 2026, https://www.kaggle.com/datasets/sagieppel/labpics-chemistry-labpics-medical
20. Comprehensive image dataset for enhancing object detection in chemical experiments - NIH, accessed January 11, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10827390/
21. Computer Vision for Recognition of Materials and Vessels in Chemistry Lab Settings and the Vector-LabPics Data Set | ACS Central Science, accessed January 11, 2026, https://pubs.acs.org/doi/10.1021/acscentsci.0c00460
22. Hazmat Placard Image Classification Dataset - Kaggle, accessed January 11, 2026, https://www.kaggle.com/datasets/anmatngu/hazmat-placard-image-dataset

23. GHS pictograms - UNECE, accessed January 11, 2026, https://unece.org/transport/dangerous-goods/ghs-pictograms
24. GHS Object Detection Dataset by fireljj - Roboflow Universe, accessed January 11, 2026, https://universe.roboflow.com/fireljj/ghs-hze4d
25. Biomedical Waste Classification Images Dataset - Kaggle, accessed January 11, 2026, https://www.kaggle.com/datasets/mario78/biomedical-waste-classification-images-dataset
26. nikhilvenkatkumsetty/TrashBox: Dataset of trash objects for waste classification and detection - GitHub, accessed January 11, 2026, https://github.com/nikhilvenkatkumsetty/TrashBox
27. p48 Object Detection Dataset by pfand48 - Roboflow Universe, accessed January 11, 2026, https://universe.roboflow.com/pfand48/p48
28. bank bottle Object Detection Model by zz - Roboflow Universe, accessed January 11, 2026, https://universe.roboflow.com/zz-77ljl/bank-bottle
29. taeyoungYoo/rvm-dataset - GitHub, accessed January 11, 2026, https://github.com/taeyoungYoo/rvm-dataset
30. Reverse vending machine Computer Vision Model, accessed January 11, 2026, https://universe.roboflow.com/reverse-vending-machine-lqerh/reverse-vending-machine-ogew0
31. WaDaBa, accessed January 11, 2026, https://wadaba.pcz.pl/
32. NIR-MFCO dataset: Near-infrared-based false-color images of post-consumer plastics at different material flow compositions and material flow presentations | Faculty 5 | RWTH Aachen University | EN, accessed January 11, 2026, https://www.fb5.rwth-aachen.de/cms/georessourcen/forschung/publikationen/publikationen/~bbvzxu/details/?file=954195&lidx=1
33. Licensing of sales packaging - Grüner Punkt, accessed January 11, 2026, https://www.gruener-punkt.de/en/packaging-licensing/sales-packaging
34. 3+ Thousand German Recycling Royalty-Free Images, Stock Photos & Pictures, accessed January 11, 2026, https://www.shutterstock.com/search/german-recycling
35. (PDF) A Smart Recycling Bin for Waste Classification - ResearchGate, accessed January 11, 2026, https://www.researchgate.net/publication/338601590_A_Smart_Recycling_Bin_for_Waste_Classification
36. Construction and Demolition Waste Dataset and YOLOv11-L Model ..., accessed January 11, 2026, https://zenodo.org/records/17447590
37. Solid Waste Detection Using Enhanced YOLOv8 Lightweight Convolutional Neural Networks - MDPI, accessed January 11, 2026, https://www.mdpi.com/2227-7390/12/14/2185
38. kendrickfff/waste-classification-yolov8-ken - Hugging Face, accessed January 11, 2026, https://huggingface.co/kendrickfff/waste-classification-yolov8-ken